

Modelling Returns and Volatility in the Swedish Stock Market with ARIMA and GARCH.

1. Introduction

This paper studies the Swedish stock market by modelling and forecasting the return and volatility of the OMX Stockholm 30 (OMXS30) index, consisting of some of the largest companies in Sweden. Studying this has important implications for areas such as risk and portfolio management, derivatives pricing and other structured products. Furthermore, being able to model and forecast volatility, which for example has been shown to cluster during times of instability (Mandelbrot, 1963), is due to this important for policymakers and investors.

The models applied in the study are ARIMA (Autoregressive Integrated Moving Average) and GARCH (Generalized Autoregressive Conditional Heteroskedasticity). The ARIMA is fitted to the log return series to model the past returns and forecast the future, it is then estimated jointly with the GARCH to capture the volatility of returns. The study looks at whether an ARIMA model can describe the returns of the OMXS30. And if the residuals have ARCH effects (volatility clustering), can GARCH capture these?

2. Literature Review

Mandelbrot (1963) observed that large (small) price changes are often followed by large (small) price changes, which is now known as volatility clustering. It has also been shown that stock return distributions have heavy tails (Fama, 1965), meaning extreme market returns are more usual than expected in a normal distribution. This study will analyse whether this can also be seen in the OMXS30 data, and if so, take it into account with the models and approach used.

Some newer empirical studies, such as Vo and Ślepaczuk (2022), have applied ARIMA-GARCH frameworks for the S&P 500 and found that jointly estimating the two models has higher forecasting power than the ARIMA alone. Aliyev et al. (2020), who model the volatility for the Nasdaq-100 Index for 2000-2019 find that there exists volatility clustering. In addition, they also find that there is a leverage effect, meaning negative returns of the same magnitude as positive returns have a higher impact on volatility.

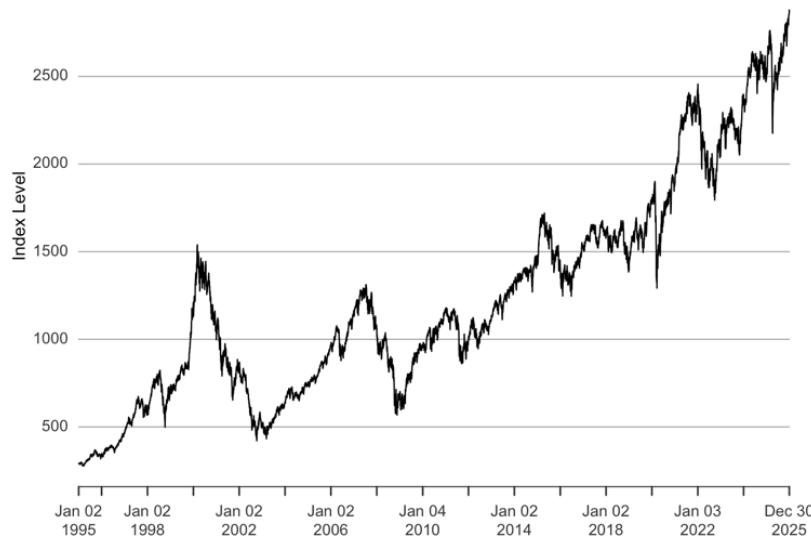
3. Data

This study uses unadjusted daily closing prices for the OMXS30 index over the period: 1995-2025, mainly because it is a recent period, seen as sufficiently long and includes several distinct market events such as the dot-com bubble (2000-2002), the 2008 financial crisis and the 2020 stock market crash during COVID-19.

The data has been obtained from Yahoo Finance (n.d.), Investing.com (n.d.) and directly from Nasdaq (n.d.). Yahoo Finance only had data available from around 2008, therefore Investing.com was used for data before this date. Furthermore, the data from Yahoo Finance

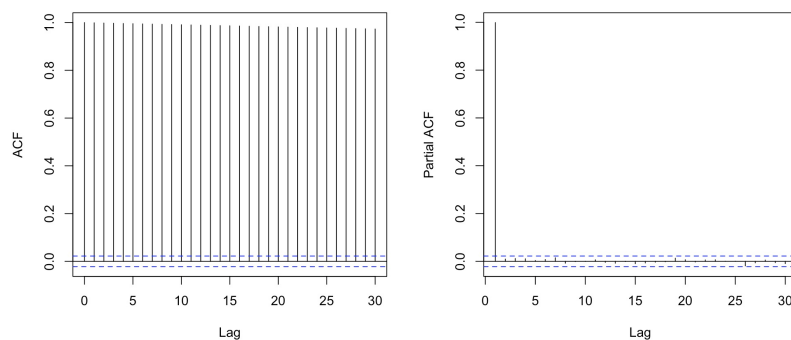
had some missing values for a total of six trading days. Two of these trading days had data from Nasdaq, hence these were used and the remaining four trading days were omitted. Additionally, some of the data entries were stored as characters with thousands separators, which were converted to numeric format before the analysis. The total number of observations after the data preparation was 7783.

Figure 1. OMXS30 Index Level – in levels. 1995-2025.



Note: The figure plots daily unadjusted closing prices of the OMXS30 Index from January 2, 1995, to December 30, 2025. (Investing.com, n.d.; Nasdaq, n.d.; Yahoo Finance, n.d.) N = 7783 observations.

Figure 2. ACF and PACF of the log price level.



Note: Sample autocorrelation function (ACF, left) and partial autocorrelation function (PACF, right) of the log price level of the OMXS30 for 0-30 lags. Dashed blue lines denote 95% confidence intervals under the null hypothesis of no autocorrelation.

visual interpretation, we conclude that the time series is non-stationary. Schwert (1989) outlines the following guideline for choosing the maximum number of lags in the ADF: $(12 \times (T/100)^{1/4})$, where T is the number of observations. In this study it corresponds to approximately 35 lags.

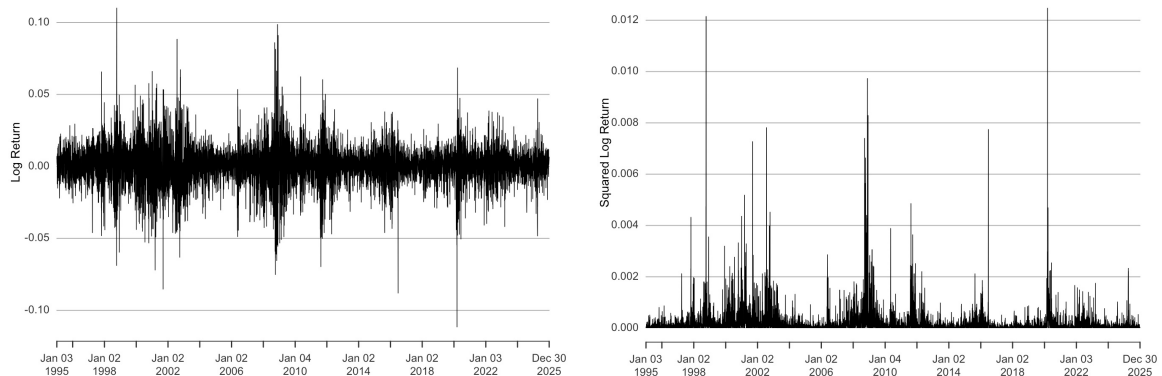
The solution for this is to take the first difference, and when taking the first difference of log prices, this yields the log returns and can make the series stationary. Figure 3 shows the log

Figure 1 shows the OMXS30 over the time period. It is visually clear that there is an upward trend and that the time series is non-stationary. For further investigation, the log price level is computed and the ACF and PACF can be seen in Figure 2.

We see strong persistence and slow decay of the ACF, indicating non-stationarity. An Augmented Dickey Fuller (ADF) test is conducted using a maximum of 35 lags based on Schwert (1989), which yields a test statistic of -1.7586, meaning we fail to reject the null hypothesis. Furthermore, a KPSS test is conducted, and with a p-value = 0.01 we reject the null hypothesis of stationarity. Based on these tests, the ACF/PACF and the

returns of the OMXS30. Furthermore, the squared log returns are computed in the same figure; the reason for this is to see if there might be some potential volatility clustering, hence a motivation for also using the GARCH model.

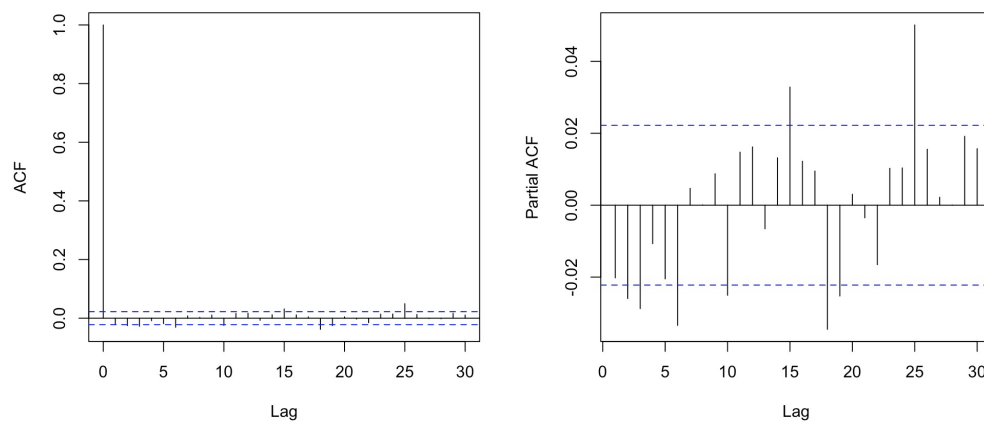
Figure 3. OMXS30 Daily Log Returns and Squared Log Returns. 1995-2025.



Note: Daily log returns (left) and squared log returns (right) of the OMXS30 from January 3, 1995 to December 30, 2025. $N = 7782$ observations. Log returns computed as $r_t = \ln(P_t/P_{t-1})$.

Visually, in Figure 3, the time series looks more stationary and the mean seems more constant than before the differencing. In addition, there might be some volatility clustering as seen with the squared log returns, with periods of high volatility and some with lower volatility. Below the ACF and PACF on the log returns are displayed.

Figure 4. ACF and PACF of Log Returns.



Note: Sample autocorrelation function (ACF, left) and partial autocorrelation function (PACF, right) of the daily log returns of the OMXS30 for 0-30 lags. Dashed blue lines denote 95% confidence intervals under the null hypothesis of no autocorrelation.

The ACF now cuts off directly, with few significant spikes and the PACF has low magnitude, even if multiple spikes are significant. Hence, these also point towards that the first differencing has led to a stationary time series.

Additionally, an ADF test is done on the differenced data, the test statistic is -14.35 and with a 1% critical value of -3.43; the null hypothesis of non-stationarity is rejected. Also, a KPSS test shows a p-value of above 0.1 meaning the null hypothesis of stationarity fails to be

rejected at the 5% level. Based on these tests, the ACF/PACF and the visual interpretation, it is concluded that the series is stationary. Below are the descriptive statistics for the series.

Table 1. Descriptive statistics - Log Returns.

Statistic	Value
Mean	0.000296
Std. Dev.	0.013946
Minimum	-0.111726
Maximum	0.110228
Skewness	-0.061
Excess kurtosis	1.6083
Observations	7782

Note: Log returns are computed as $\ln(P_t) - \ln(P_{t-1}) = \ln(P_t/P_{t-1})$. The sample covers daily closing prices for OMXS30 from January 1995 to December 2025.

The mean daily log return is 0.0296%, which annualised is around 7.46% (0.000296×252). The standard deviation of around 1.39% per day is around 22% annually ($0.0139 \times \sqrt{252}$). These numbers are consistent with what is found in the literature for other major stock indices, such as the Dow Jones Industrial Index from Engle and Patton (2001). The minimum and maximum returns of around -11% and +11% respectively show the drastic events during the sample period, such as COVID-19. Moreover, the distribution seems to have a slight negative skewness and excess kurtosis, also shown by Fama (1965) for

equity returns. The number of observations drops from 7783 to 7782, due to the differencing.

4. Methodology

4.1 - ARIMA

An ARIMA(p,d,q) model is used in this study to model and study the time series. As stationarity of the time series has been shown in Section 3, the integration order is set to 0 (I(0)). Hence, the general form of this model, also called ARMA(p,q), is the following:

$$y_t = a_0 + \sum_{i=1}^p a_i y_{t-i} + \sum_{j=1}^q \beta_j \varepsilon_{t-j} + \varepsilon_t, \quad \varepsilon_t \sim i.i.d. (0, \sigma^2)$$

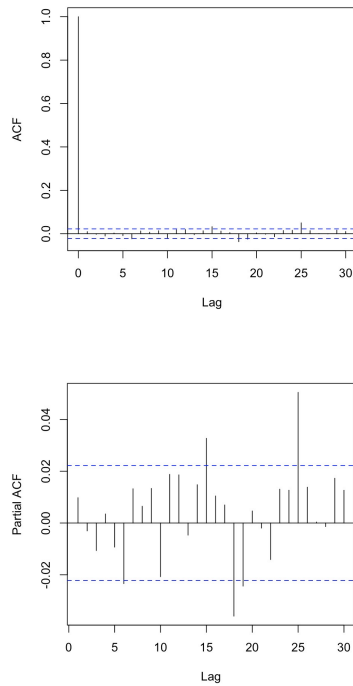
where a_0 is the constant mean (intercept), a_i are the autoregressive (AR) coefficients, β_j are the moving average (MA) coefficients and ε_t is a white noise process. For the model to be stationary and invertible, the roots of the AR and MA polynomials must be outside the unit circle.

To choose a model, the ACF and PACF (Figure 4) can be inspected, however these alone do not seem to give a clear indication of the correct model – it does potentially look like an ARMA(0,0). This would mean that the returns follow a white noise process, meaning returns are unpredictable, which is consistent with the efficient market hypothesis (Fama, 1970).

Due to this unclarity, subsequent steps are taken with the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) to select the model specification. These information criteria measure the goodness of fit, with the sum of squared residuals (SSR) and penalise based on how many parameters are included. Hence, the model specification with the lowest information criteria is preferred, however it is important not to cross-compare the criteria, due to different structures. Using the BIC can result in a more parsimonious model, whereas the AIC can be more complex.

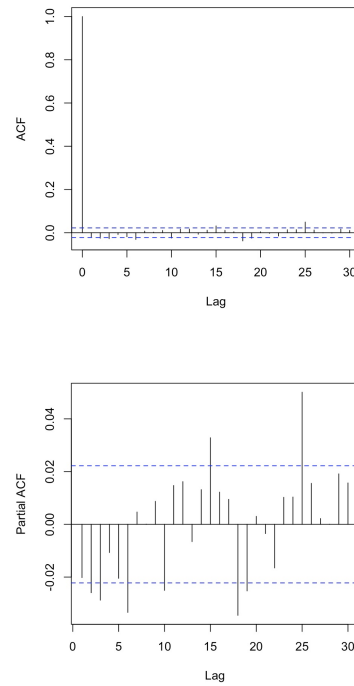
For this study, using the BIC results in an ARMA(0,0), which is expected from looking at the ACF/PACF (Figure 4) and because of parsimony. However, using the AIC for model selection yields an ARMA(1,1). Therefore, to choose a model specification, the residuals are inspected through the ACF and PACF for both models below.

Figure 5. ACF and PACF of residuals from the model selected using AIC.



Note: Sample autocorrelation function (ACF, top) and partial autocorrelation function (PACF, bottom) of the residuals from the ARIMA model selected by AIC, computed for 0-30 lags. Dashed blue lines show 95% confidence intervals under the null of no autocorrelation.

Figure 6. ACF and PACF of residuals from the model selected using BIC.



Note: Sample autocorrelation function (ACF, top) and partial autocorrelation function (PACF, bottom) of the residuals from the ARIMA model selected by BIC, computed for 0-30 lags. Dashed blue lines show 95% confidence intervals under the null of no autocorrelation.

The ACF/PACF seems to be fairly similar for the residuals for both model specifications, with some significant spikes. A Ljung-Box test is then applied to the residuals of each candidate model, using ten lags. The number of lags considered is based on $\ln(\text{number of observations})$ (Tsay, 2005, p. 33) which here is approximately ten. From the model specification using the AIC, the p-value is around 0.21 and for using the BIC the p-value is around 0.0005. Hence, the null hypothesis of no autocorrelation in the residuals is rejected when using the BIC, but not when using the AIC. Therefore, it is concluded that the residuals in the model specification using the AIC behave like white noise. Based on

this, the final model specification used is ARMA(1,1). However, it is important to see that if the number of lags considered in the Ljung-Box test were higher (such as with 15 or 20 lags considered), the null hypothesis would also be rejected using the AIC, which can also, to some extent, be seen in Figure 5. Furthermore, we now have a less parsimonious model, however because there is more indication that the residuals are white noise – the ARMA(1,1) model is chosen, and the coefficients and standard errors can be seen below.

$$y_t = \frac{0.0003}{(0.0001)} + \frac{0.7468}{(0.0673)} y_{t-1} - \frac{0.7790}{(0.0633)} \varepsilon_{t-1} + \varepsilon_t$$

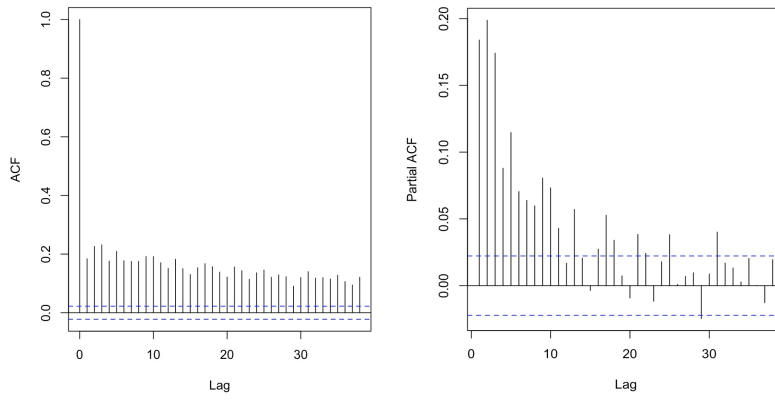
Based on the standard errors (in parentheses), all coefficients are statistically significant. Furthermore, the stationarity and invertibility conditions are satisfied as the AR and MA coefficients lie within the unit circle. However, we also see that the coefficients almost cancel each other out, meaning a common factor problem, where there is a risk of redundant

parameters and that in reality we are just overcomplicating the model and that the more parsimonious ARMA(0,0) would be better. We acknowledge this, but at the same time, based on the Ljung-Box test, the residuals in the ARMA(1,1) behaved closer to white noise, hence we stick with the ARMA(1,1).

4.2 – GARCH

As already seen in Figure 3, there might be some volatility clustering, which can't be captured by an ARIMA model alone, as it assumes constant volatility. To further test this, the ACF and PACF of the squared residuals from the ARMA(1,1) model are inspected. Also, an ARCH-LM test is conducted.

Figure 7. ACF and PACF of Squared Residuals from ARMA(1,1).



Note: Sample autocorrelation function (ACF, left) and partial autocorrelation function (PACF, right) of the squared residuals from the ARMA(1,1) model fitted to OMXS30 daily log returns. Dashed blue lines show 95% confidence intervals under the null of no autocorrelation.

It is clear that there is much more persistence and significance in the ACF/PACF. Also, the ARCH-LM test using 10 lags, gives a p-value < 0.001 and using even fewer lags still yields a significant result. Meaning the null hypothesis can be rejected and we can conclude that there are ARCH effects, which is a motivation for

also including the GARCH model.

The specific model specification used in this study is a GARCH(1,1), based on Hansen and Lunde (2005) who find that this specification is not outperformed by more sophisticated models. Hansen and Lunde specifically observe this for the analysis of exchange rates, meaning that it could differ when applied to a stock index. They also show that there are better specifications for single stocks, however we still adopt it due to its parsimony and will later also inspect if it is adequate for the data in this study. In addition, Srinivasan (2011) shows that for the S&P 500 Index, the simple GARCH(1,1) outperforms more complex GARCH models in forecasting.

Now, the GARCH(1,1) model assumes that the error ε_t is $\varepsilon_t = \sigma_t z_t$, $z_t \sim N(0,1)$ and σ_t is the conditional standard deviation and the conditional variance in this specification is:

$$\sigma_t^2 = a_0 + a_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where a_0 is the variance intercept, a_1 captures how today's variance is affected by last period's shock (ARCH term) and β_1 captures how much the effect of the previous variance persists (GARCH term). Note that both a_1 and β_1 must be non-negative, as the variance cannot be negative, this is one of the reasons why Maximum Likelihood is used for

estimation. Furthermore, for stationarity: $a_1 + \beta_1 < 1$ and this creates a mean-reverting volatility process.

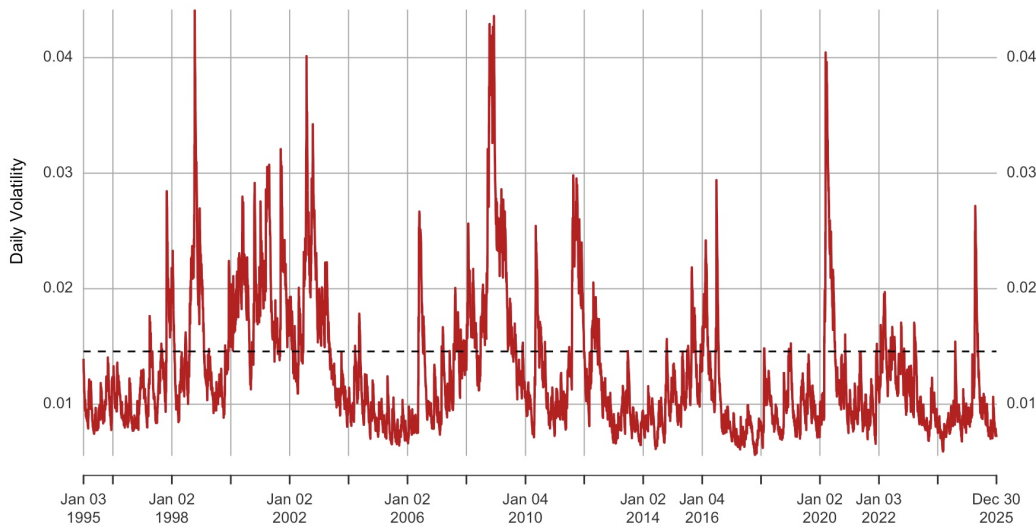
Then a joint estimation is done with Maximum Likelihood with the ARMA(1,1) as the mean model and GARCH(1,1) as the variance model. In addition, given the information about excess kurtosis in Table 1, the distribution model for the errors is a standardised t-distribution, instead of a standard normal distribution, to take into account the fatter tails following Bollerslev (1987). The estimation yields the following model for the conditional variance:

$$\sigma_t^2 = \frac{1.8631 \times 10^{-6}}{(1.0392 \times 10^{-6})} + \frac{0.0857}{(0.0125)} \varepsilon_{t-1}^2 + \frac{0.9055}{(0.0134)} \sigma_{t-1}^2$$

The coefficients are all statistically significant on the 1% level (based on the standard errors in parentheses), except for the intercept which is statistically significant on the 10% level. $a_1 + \beta_1 = 0.991$, meaning it is less than, but very close to 1. Hence, there is stationarity and highly persistent conditional variance, meaning mean reversion of volatility happens slowly. Furthermore, the β_1 -coefficient is much greater than the a_1 -coefficient, meaning that the conditional variance is mainly driven by the persistence of previous variance rather than previous shocks or previous extreme returns. This result is similar to what Engle and Patton (2001) showed for the Dow Jones Industrial Index, where a_1 is estimated to be 0.0399 and β_1 is estimated to be 0.9505.

Additionally, Figure 8 below shows the fitted conditional volatility of the OMXS30. In the plot, volatility clustering is clearly visible. For example, at the beginning of the 2000s, when the dot-com bubble burst or the 2008-2009 financial crisis, volatility was at an elevated level above the mean for multiple months. A mean reversion can also be seen, where the volatility returns to the baseline; the unconditional daily volatility of 1.46% after periods of higher or lower volatility.

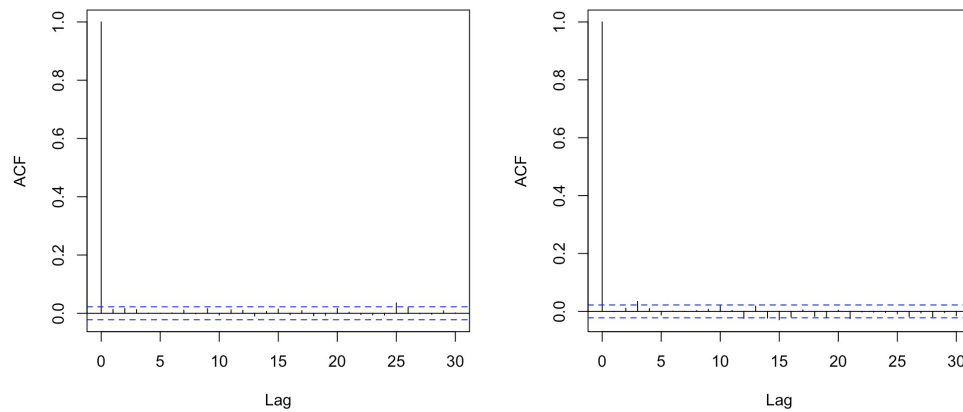
Figure 8. Fitted conditional volatility OMXS30. 1995-2025.



Note: Fitted conditional volatility (red line) from the GARCH(1,1) model fitted to OMXS30 daily log returns from 3 January, 1995, to 30 December, 2025. The dashed black line represents the unconditional volatility implied by the GARCH(1,1), computed as: $\sqrt{a_0/1 - a_1 - \beta_1}$.

To control whether the specified model is adequate the standardised residuals (ε_t/σ_t) from the GARCH(1,1) are inspected. If there is serial correlation, then the mean model (the ARMA(1,1)) is not correctly specified. We inspect the ACF for this and conduct a Ljung-Box test (with ten lags as before):

Figure 9. ACF of standardised residuals and squared residuals from GARCH(1,1).



Note: Sample autocorrelation function of the standardised residuals (left) and squared standardised residuals (right) from the GARCH(1,1) model fitted to OMXS30 daily log returns. Standardised residuals are defined as: ε_t/σ_t . Dashed blue lines show 95% confidence intervals under the null of no autocorrelation.

The Ljung-Box test yields a p-value = 0.5994. Based on this test and looking at the ACF, we conclude that there is not a significant amount of serial correlation in the standardised residuals, hence the mean equation is properly specified.

Furthermore, the squared standardised residuals are inspected with the ACF in Figure 9, a Ljung-Box (with ten lags as before) and an ARCH-LM to test for remaining ARCH/GARCH effects. Visually in the plot, we see that only a few spikes are significant and the Ljung-Box test yields a p-value = 0.1132, meaning we cannot reject the null hypothesis of no serial correlation. The ARCH-LM test yields a p-value = 0.1125, meaning that we can conclude that the variance equation is specified correctly and that the GARCH(1,1) is adequate for the data in this study.

5. Results

This is the estimated ARMA(1,1) model outlined in Section 4:

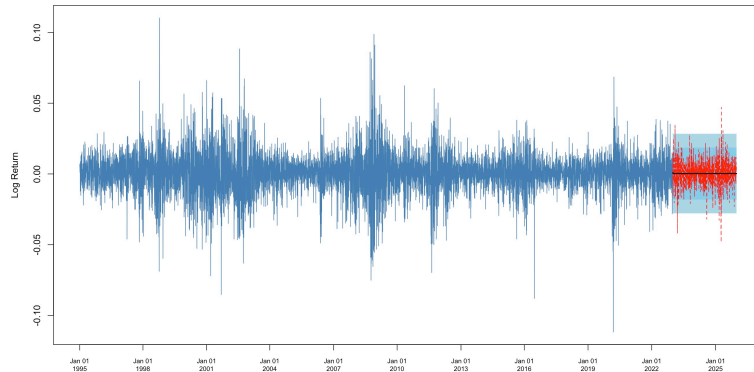
Table 2. ARMA(1,1) parameter estimates.

Parameter	Estimate	Standard error	p-value
Intercept	0.0003**	0.0001	p = 0.029
AR(1)	0.7468***	0.0673	p < 0.001
MA(1)	-0.7790***	0.0633	p < 0.001

Note: Estimated coefficients from the ARMA(1,1) model fitted to OMXS30 daily log returns. *** denotes significance at the 1% level, ** at the 5% level.

This model can then be used for forecasting, which in this study is done by splitting the data into a training set and a test set. The forecast will be 756 trading days (three trading years) and hence the test set will be around 10% of observations. The other around 90% will be used to fit the model; this is done to be able to evaluate the forecast. The forecast can be seen in the plot below:

Figure 10. $ARMA(1,1)$ – 3 trading year forecast.



Note: In-sample log returns of the OMXS30 (blue) together with a 756-trading-day (≈ 3 trading-year) out-of-sample forecast (black, conditional mean) from $ARMA(1,1)$ compared to realised log returns (red) over the forecast horizon. Shaded bands represent the 95% confidence interval (darker blue) and 80% confidence interval (lighter blue).

Figure 10 shows the forecast. As the process is stationary, the best prediction that can be done with an ARIMA model in the long run is essentially the mean. This can also be seen here, as the forecast converges towards the unconditional mean of returns. Furthermore, the confidence intervals are wide and constant, and realised returns cross outside of them multiple times, reflecting why a GARCH model is important. In

addition, the mean absolute error can be computed which yields: $MAE = 0.006929$, meaning on average the forecast misses the actual realised return with around 0.7 percentage points per day. The root mean squared error is: $RMSE = 0.009389$ and as it is around 1.36 times greater than MAE, which indicates heavier tails. Furthermore, the standard deviation of the test sample is 0.009394, which is essentially identical to the RMSE, meaning the ARIMA forecast performs basically no better than just predicting the mean.

In Table 3 is the GARCH(1,1) model outlined in Section 4.

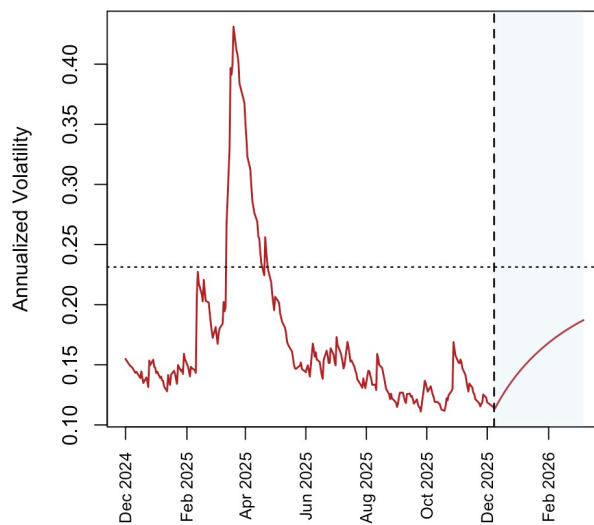
Table 3. Joint estimation of $ARMA(1,1)$ -GARCH(1,1) coefficients.

Parameter	Estimate	Standard error	p-value
- ARIMA			
Intercept	0.0008 ***	0.0001	$p < 0.001$
AR(1)	0.8268 ***	0.0677	$p < 0.001$
MA(1)	-0.8532 ***	0.0628	$p < 0.001$
- GARCH			
Intercept	1.8631×10^{-6} *	1.0392×10^{-6}	$p = 0.073$
α_1	0.0857 ***	0.0125	$p < 0.001$
β_1	0.9055 ***	0.0134	$p < 0.001$
Shape	9.74 ***	0.88	$p < 0.001$

Note: The model is estimated by MLE under the t-distribution. The ARIMA part describes the conditional mean, and the GARCH part describes the conditional variance. The shape parameter is the degrees of freedom of the t-distribution. *** denotes significance at the 1% level, * at the 10% level.

We see here that the ARMA(1,1) mean equation now has slightly different parameters, because of the joint estimation which accounts for the conditional variance. However, it is still stationary, invertible, the magnitude is similar and the signs are the same compared to estimating ARMA alone. This can then be used in volatility forecasting and the full sample is used in Figure 11 for a 90-trading-day-ahead forecast. In the plot, only the last year of the conditional volatility will be shown together with the forecast, mainly to make it more visual (however, the full sample was used for the forecast).

Figure 11. 90-trading-day ahead forecast.



Note: In-sample fitted conditional volatility (red, before vertical dashed line) and 90-trading-day out-of-sample volatility forecast (red, after vertical dashed line and shaded region) from ARMA(1,1)-GARCH(1,1) model fitted to OMXS30 daily log returns. The forecast is estimated on the full sample (January 1995 to December 2025); but only the last 12 months are displayed for visual clarity. Volatility is annualised by multiplying the daily conditional volatility by $\sqrt{252}$. The dashed horizontal black line represents the unconditional volatility implied by the GARCH(1,1), computed as: $\sqrt{a_0 / (1 - a_1 - \beta_1)}$.

We see that the forecast starts to revert to the unconditional annualised volatility of around 23% (the dashed horizontal line). Even after 90 days, it still hasn't reached the unconditional volatility, which shows the persistence. This is governed by $a_1 + \beta_1 = 0.991$ and it implies a half-life of volatility of 76.7 trading days ($\ln(0.5)/\ln(0.991)$). Meaning volatility decays by 50% over 76.7 trading days, or around 4 months. And we see this by our forecast being slightly above the midpoint between the start of the forecast and the unconditional volatility. So we see that periods of low volatility, as in this example, are not expected to persist but eventually return to the mean.

6. Conclusion

In this paper, the OMXS30 index was analysed and we saw that it had similar dynamics to other equity indices. Firstly, the non-stationary log prices become stationary after differencing, the mean daily log return

is positive, but close to zero, with excess kurtosis and volatility clustering. An ARMA(1,1) model was selected and we saw that the forecast quickly went toward the unconditional mean, as expected due to stationarity. Furthermore, a GARCH(1,1) model was also used and showed high volatility persistence for the OMXS30 index, which has also been shown for other major stock indices. Extensions to this study could be to use asymmetric extensions of the GARCH model, such as the EGARCH and GJR-GARCH, to allow for negative shocks to raise volatility more than positive shocks, also known as the leverage effect (Christie, 1982). A further extension would be to formally evaluate the out-of-sample volatility forecast, where we could then compare different GARCH models. This could be for example by using a proxy (like high-frequency intra-trading-day data) in a Mincer-Zarnowitz regression, since the conditional volatility is not directly observable, as Hansen and Lunde (2005) did.

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